

Introduction to Cryptography

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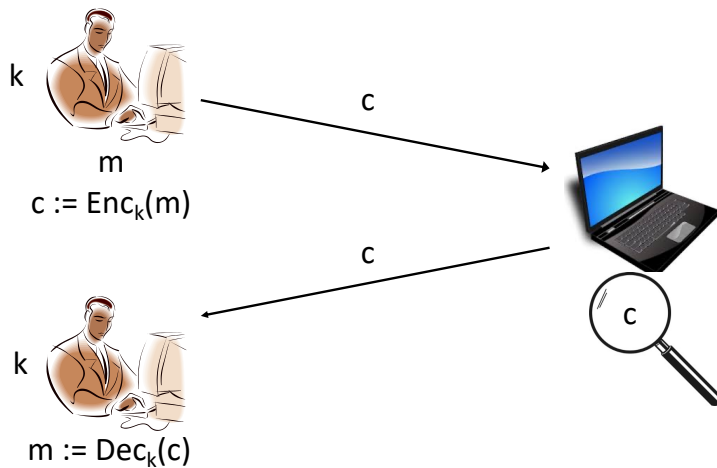
August 29, 2026

Classical cryptography

Why start with classical cryptography?

- Historically, cryptography focused on *secrecy of communication*.
- Communicating parties shared a secret *key* in advance.
 - Private-key, secret-key, shared-key, or symmetric-key cryptography
- These schemes introduce our notation and the attacker's mindset.
- They also show why unprincipled design is more dangerous than it appears.

Private-key encryption



Private-key encryption, formally

- A private-key encryption scheme has a message space $\mathcal{M}$ and three algorithms:
 - Gen outputs a key $k \in \mathcal{K}$.
 - $\text{Enc}_k(m)$ takes $m \in \mathcal{M}$ and outputs ciphertext c .
 - $\text{Dec}_k(c)$ outputs a message or an error.
- **Correctness:** for every $m \in \mathcal{M}$ and every key output by Gen,

$$\text{Dec}_k(\text{Enc}_k(m)) = m.$$

Kerckhoffs's principle

- The attacker knows the encryption scheme; only the key is secret.
- Keys are easier to protect and replace than algorithms.
- Public designs can be standardized and scrutinized before deployment.
- Therefore, security must not depend on hiding how the scheme works.

The shift cipher

- Identify the letters with $\{0, \dots, 25\}$: $a = 0, b = 1, \dots, z = 25$.
- Choose a key $k \in \{0, \dots, 25\}$.
- Encrypt by shifting every plaintext letter forward by k , wrapping around at 26.
- Decrypt by shifting backward by k .

helloworld $\xrightarrow{k=2}$ jgnnqyqtnf

Modular arithmetic

- $[x \bmod N]$ is the unique remainder in $\{0, \dots, N - 1\}$.
- Two integers are congruent modulo N when their difference is divisible by N :

$$x \equiv y \pmod{N} \iff N \mid (x - y).$$

- For example, $35 \equiv 5 \pmod{10}$ and $[35 \bmod 10] = 5$.
- Modular arithmetic gives us wraparound without special cases.

The shift cipher, formally

- $\mathcal{M}$ is the set of strings over the lowercase English alphabet.
- Gen chooses uniform $k \in \{0, \dots, 25\}$.
- For $m = m_1 \cdots m_t$,

$$c_i := [m_i + k \bmod 26].$$

- Decryption computes $m_i := [c_i - k \bmod 26]$.
- Correctness follows because the two shifts cancel modulo 26.

Is the shift cipher secure?

- No: there are only 26 possible keys.
- Given a ciphertext, try every key and inspect the candidate plaintexts.
- For a sufficiently long English message, usually only one candidate makes sense.
- This is an *exhaustive-search* or *brute-force* attack.

uryybjbeyq $\xrightarrow{\text{try all keys}}$ helloworld

Quiz 1

- Using the English-language shift cipher, what is the encryption of “good” using the key ‘b’?
 - XYYD
 - HPPE
 - QRST
 - GNNE

Quiz 1 – answer

- The correct answer is HPPE
- Using the key 'b', we shift each letter of “good” forward by 1:
 - $g \rightarrow h$
 - $o \rightarrow p$
 - $o \rightarrow p$
 - $d \rightarrow e$

Quiz 2

- Using the English-language shift cipher, which of the following plaintexts could correspond to ciphertext “AZC”?
 - can
 - bad
 - dog
 - run

Bytes, hexadecimal, and ASCII

- Arbitrary data is represented as bytes; one byte is 8 bits.
- A byte is written using two hexadecimal digits: 0x00 through 0xFF.
 - $0x41 = 0100\ 0001_2 = 65$
 - $0xAF = 1010\ 1111_2 = 175$
- ASCII assigns characters to byte values: for example, A is 0x41 and a is 0x61.
- Printable ASCII occupies 0x20–0x7E, which gives an attacker useful structure.

Byte-wise shift cipher – extending the shift cipher

- $\mathcal{M}$ is the set of byte strings; choose $k \in \{0x00, \dots, 0xFF\}$.
- Replace modular addition with bitwise XOR:

$$c_i := m_i \oplus k, \quad m_i := c_i \oplus k.$$

- Correctness follows from $(x \oplus k) \oplus k = x$.
- The construction works on arbitrary data, but still has only 256 keys.

The lesson from exhaustive search

- Trying all 256 byte keys is easy.
- A key space must be large enough to make exhaustive search impractical.
- A large key space is *necessary*, but not sufficient for security.
 - A scheme may leak structure that enables a faster attack.
- The Vigenère cipher illustrates exactly this failure.

The Vigenère cipher

- The key is now a *string*, not just a character
- To encrypt, shift each character in the plaintext by the amount dictated by the next character of the key
 - Wrap around in the key as needed
- Decryption just reverses the process
- An example
 - Plaintext: tellhimaboutme
 - Key: cafecafecafeca
 - Ciphertext: veqpjiredozxoe

The Vigenère cipher – analysis

- Size of key space?
 - If keys are 14-character strings over the English alphabet, then key space has size $26^{14} \approx 2^{66}$
 - If variable length keys, even more...
 - Brute-force search infeasible
- Is the Vigenère cipher secure?
- (Believed secure for many years. . .)

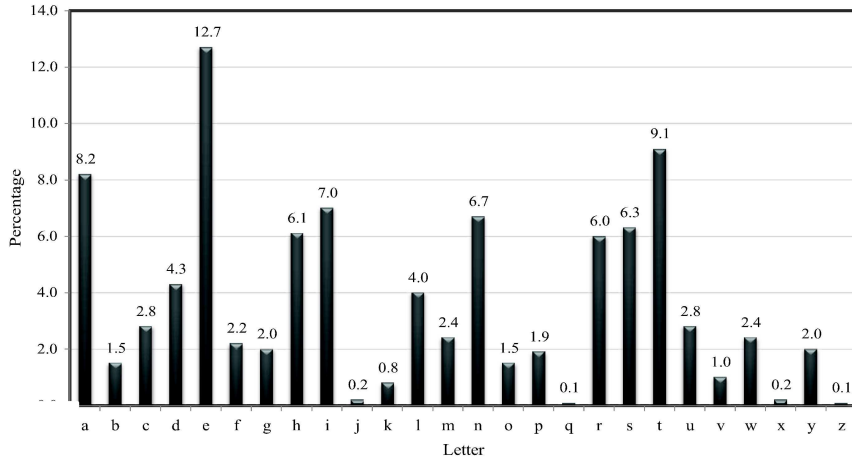
Attacking the Vigenère cipher – first try

- (Assume a 14-character key)
- Observation: every 14th character is “encrypted” using the same shift
- Looking at every 14th character is(almost) like looking at ciphertext encrypted with the shift cipher
 - Though a direct brute-force attack doesn't work...
 - Why not?

Attacking the Vigenère cipher – an example

- The ciphertext is
veqppjiredozxoeualpcmsdjuiqndnossoscdcusoakjqm xpqrhyycjqoqqodhjcciwieii
- Look at every 14th character
veqppjiredozxoe**u**alpcmsdjuiq**n**dnossoscdcuso**a**kjqm xpqrhyycjq**o**qqodhjcciwie**i**i
- Why bruteforce search does not work?
 - How to verify the correctness of a candidate key?

Using plaintext letter frequencies



Attacking the Vigenère cipher – a better approach

- Look at every 14th character of the ciphertext, starting with the first
 - Call this a “stream”
- Let α be the most common character appearing in this stream
- Most likely, α corresponds to the most common plaintext character (i.e., 'e')
 - Guess that the first character of the key is α - 'e'
- Repeat for all other positions
- This is somewhat haphazard... and does not use all the information available

An even better attack

- Let p_i ($0 \leq i \leq 25$) denote the frequency of the i^{th} English letter in normal English plaintext
 - One can compute that $\sum_i p_i^2 \approx 0.065$
- Let q_i denote the observed frequency of the i^{th} letter in a given stream of the ciphertext
- If the shift for that stream is j , expect $q_{i+j} \approx p_i$ for all i
 - So expect $\sum_i p_i q_{i+j} \approx 0.065$
- Test for every value of j to find the right one
 - Repeat for each stream

Finding the key length

- The previous attack assumes we know the key length
 - What if we don't?
- Note: can always try the previous attack for all possible key lengths
 - # of key lengths $\ll$ # keys
- Possible to do better!

Finding the key length

- When using the correct key length, the ciphertext frequencies $\{q_i\}$ of a stream will be *shifted versions* of the $\{p_i\}$
 - So $\sum q_i^2 \approx \sum p_i^2 \approx 0.065$
- When using an incorrect key length, expect (heuristically) that ciphertext letters are uniform
 - So $\sum q_i^2 \approx \sum (1/26)^2 = 1/26 = 0.038$
- In fact, good enough to find the key length N that maximizes $\sum q_i^2$
 - Can verify by looking at other streams. . .

Byte-wise Vigenère cipher

- The key is a string of bytes
- The plaintext is a string of bytes
- To encrypt, XOR each character in the plaintext with the next character of the key
 - Wrap around in the key as needed
- Decryption just reverses the process

Byte-wise Vigenère cipher – example

- Say plaintext is “Hello!” and key is 0xA1 2F
- “Hello!” = 0x48 65 6C 6C 6F 21
- XOR with 0xA1 2F A1 2F A1 2F
- $0x48 \oplus 0xA1$
 - $0100\ 1000 \oplus 1010\ 0001 = 1110\ 1001 = 0xE9$
- Ciphertext: 0xE9 4A CD 43 CE 0E
- How to attack this cipher? Same principles as before...

From attacks to modern cryptography

- Classical design followed a cycle: construct, attack, patch, repeat.
- A large key space did not tell us whether a construction was secure.
- Modern cryptography begins by replacing “looks secure” with:
 - a precise security definition,
 - clearly stated assumptions, and
 - a proof that the construction meets the definition under those assumptions.
- Next time: what should secrecy mean?